

React Components

Objectives

- ◆ Learn to create, customize, and conditionally display React components.

Snippets

imp→ `import moduleName from 'module'`

imn→ `import 'module'`

imd→ `import { destructuredModule } from 'module'`

imr→ `import React from 'react'`

imrd→ `import ReactDOM from 'react-dom'`

imrc→ `import React, { Component } from 'react'`

rfc

```
import React from 'react'

export default function $1() {
  return <div>$0</div>
}
```



ES7+ React/Redux/React-Native snippets v4.4.3

dsznajder | 12,480,536 | ★★★★★ (78)

Extensions for React, React-Native and Redux in JS/TS with ES7+ syntax. Customizable. Bu

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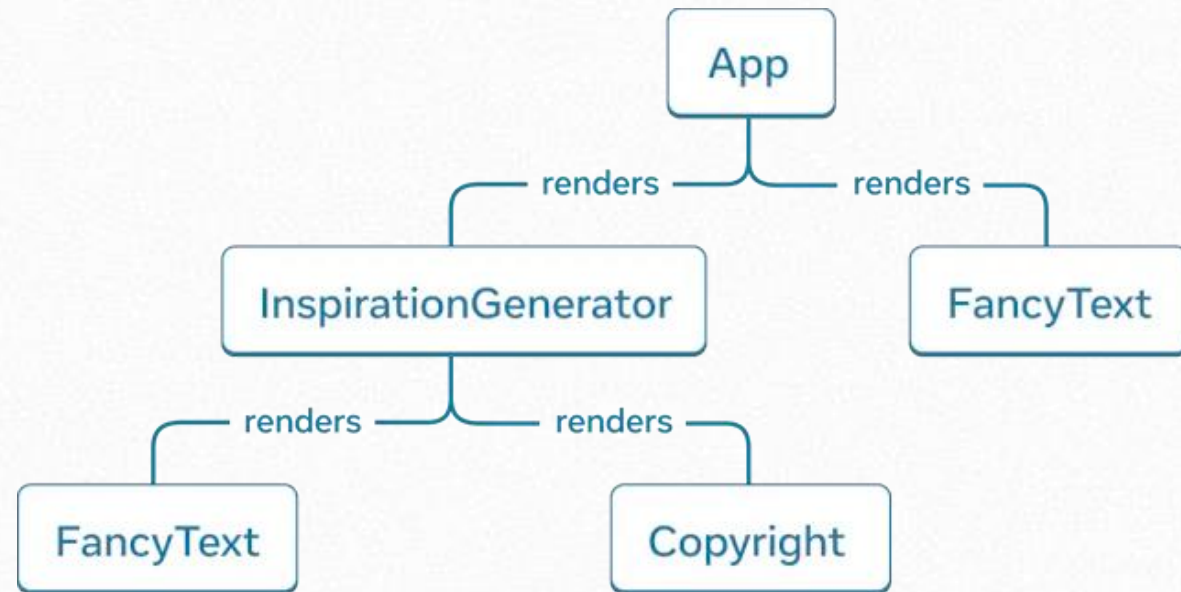
rcc

```
import React, { Component } from 'react'

export default class FileName extends Component {
  render() {
    return <div>$2</div>
  }
}
```

React Components

- ◆ *Components* enable you to split your UI into independent, reusable pieces
- ◆ *Components* are the foundation upon which you build user interfaces (UI)



Your First Component

JS HelloWorld.js U X

my-app > src > components > JS HelloWorld.js > ...

```
1  const HelloWorld = () => {
2    return (
3      <div>
4        <h2>Hello, World!</h2>
5        <p>This is your first component.</p>
6      </div>
7    );
8  };
9
10 export default HelloWorld;
```

JS App.js M X

my-app > src > JS App.js > ...

```
1  import HelloWorld from '../components/HelloWorld';
2
3  function App() {
4    return (
5      <div>
6        <h1>My React App</h1>
7        <HelloWorld />
8      </div>
9    );
10 }
11
12 export default App;
```

```
1  export default function Profile() {
2    return (
3      
7    )
8  }
```


How to build a component

1. Export the component
2. Define the function
 - Function name **must** start with *a capital letter*
 - Tags starting with lowercase letters are treated as DOM tags (Built-in components)
3. Add markup

```
1  export default function helloWorld() {  
2    return (  
3      <div>  
4        <h2>Hello, World!</h2>  
5        <p>export default testing</p>  
6      </div>  
7    );  
8  };
```

Importing and Exporting Components

- ◆ What a root component file is (App.js)
- ◆ How to import and export a component
- ◆ When to use default and named imports and exports
- ◆ How to import and export multiple components from one file

Default exports

- ◆ export default có thể xuất **một biến**, một **hàm** hoặc một **class** duy nhất từ một file.
- ◆ Một file chỉ có duy nhất một export default.
- ◆ Khi import một export default vào một file khác,
 - có thể **đặt tên tùy ý** cho giá trị được xuất.
 - không cần sử dụng cặp ngoặc nhọn {} khi import.
 - chữ HOA ký tự đầu tiên cho tên Component

Example

JS HelloWorld.js U X

my-app > src > components > JS HelloWorld.js > ...

```
1  const helloWorld = () => {  
2    return (  
3      <div>  
4        <h2>Hello, World!</h2>  
5        <p>export default testing</p>  
6      </div>  
7    );  
8  };  
9  export default helloWorld
```

JS App.js M X

my-app > src > JS App.js > ...

```
1  import Test_export_default from './components/HelloWorld';  
2  
3  function App() {  
4    return (  
5      <div>  
6        <h1>My React App</h1>  
7        <Test_export_default />  
8      </div>  
9    );  
10 }  
11  
12 export default App;
```

JS App.js M X

my-app > src > JS App.js > ...

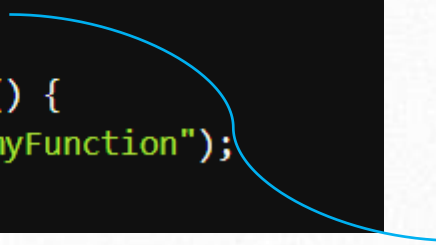
```
1  import test_export_default from './components/HelloWorld';  
2  
3  function App() {  
4    return (  
5      <div>  
6        <h1>My React App</h1>  
7        <test_export_default />  
8      </div>  
9    );  
10 }  
11  
12 export default App;
```

Named exports

- ◆ export (không default) được sử dụng khi muốn xuất **nhiều** biến, hàm, class từ một file
- ◆ Sử dụng trong cặp **{}** khi xuất và khi import.
- ◆ Sử dụng **tên chính xác** của biến, hàm, class khi import.

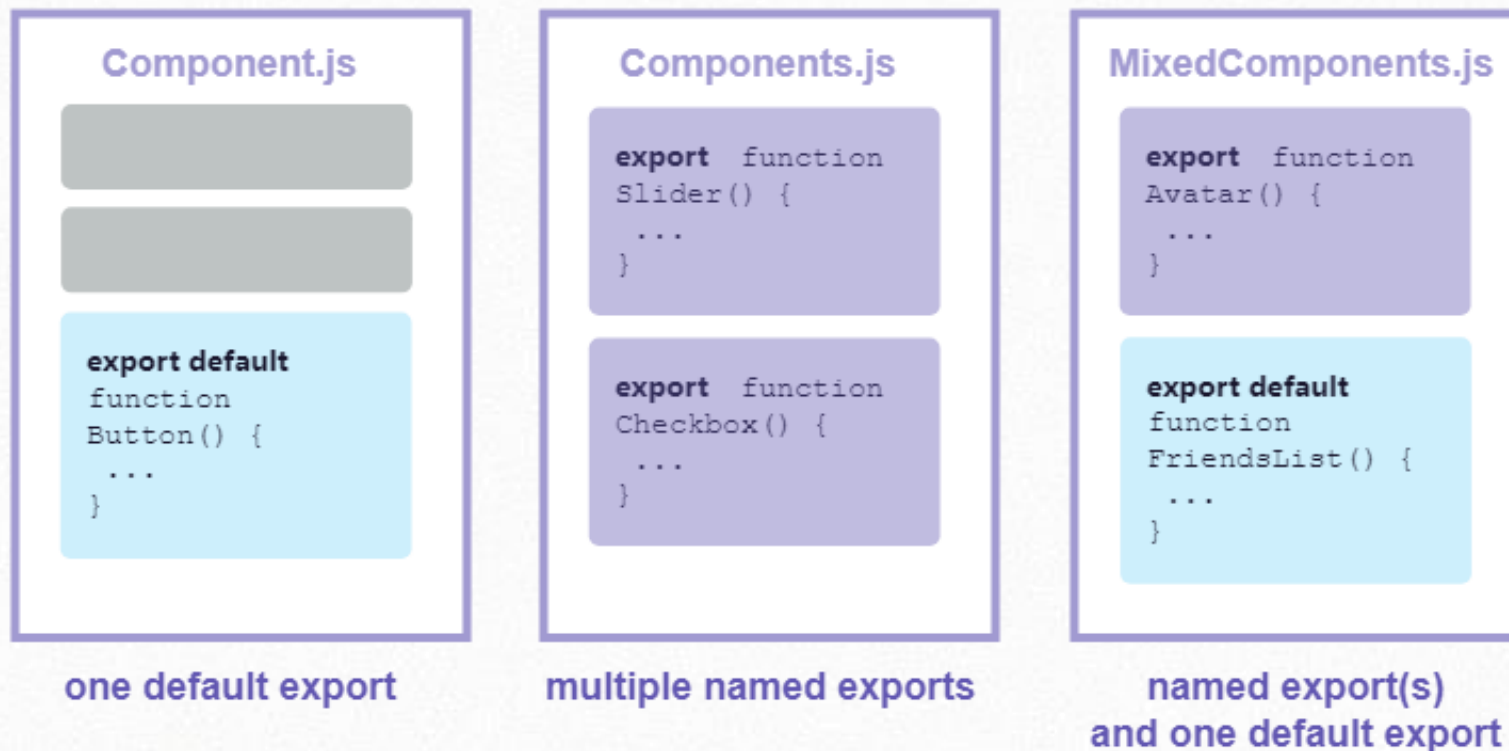
```
// File: myModule.js
export const var1 = 10;
export const var2 = 20;

export function myFunction() {
  console.log("Hello from myFunction");
}
```



```
// Importing in another file
import { var1, var2, myFunction } from './myModule.js';
console.log(var1); // Output: 10
myFunction(); // Output: Hello from myFunction
```

Default vs named exports



Syntax	Export statement	Import statement
Default	<code>export default function Button() {}</code>	<code>import Button from './Button.js';</code>
Named	<code>export function Button() {}</code>	<code>import { Button } from './Button.js';</code>

Using curly braces {}

```
1 export default function Avatar() {  
2   const avatar = 'https://i.imgur.com/7vQD0fPs.jpg';  
3   const description = 'Gregorio Y. Zara';  
4   return (  
5     <img  
6       className="avatar"  
7       src={avatar}  
8       alt={description}  
9     />  
10  );  
11 }
```

```
1 export default function TodoList() {  
2   const name = 'Gregorio Y. Zara';  
3   return (  
4     <h1>{name}'s To Do List</h1>  
5   );  
6 }
```

```
10 export default function TodoList() {  
11   return (  
12     <h1>To Do List for {formatDate(today)}</h1>  
13   );  
}
```


Object with curly braces

- ◆ Object like: { name: "Hedy Lamarr", inventions: 5 }

- ◆ Pass a JS object in JSX:

person={{ name: "Hedy Lamarr", inventions: 5 }}

- ◆ Inline CSS styles in JSX

```
1 export default function TodoList() {  
2   return (  
3     <ul style={{  
4       backgroundColor: 'black',  
5       color: 'pink'  
6     }}>
```


Kinds of components

- ◆ Function Component
- ◆ Class Component
 - extends `React.Component`
 - requires a `render()` method

```
1 export default function helloWorld() {  
2   return (  
3     <div>  
4       <h2>Hello, World!</h2>  
5       <p>export default testing</p>  
6     </div>  
7   );  
8 };
```

```
JS Student.js U X  
my-app > src > components > JS Student.js > ...  
1 import React, { Component } from 'react'  
2  
3 export default class Student extends Component {  
4   render() {  
5     return (  
6       <div>Hello Student class component!</div>  
7     )  
8   }  
9 }
```

Passing Props to a Component

```
12 export function HelloStudent({name}) {  
13   |   return (  
14   |   |   <h2>Hello {name}</h2>  
15   |   )  
16 }
```

```
18 export function HelloStudent(props) {  
19   |   return (  
20   |   |   <h2>Hello {props.name}</h2>  
21   |   )  
22 }
```

```
8  || export default function App() {  
9  |   return (  
10 ||   |   |   <HelloStudent name = "Thanh"/>  
11   |   );  
12 }
```

The diagram illustrates the flow of props from the App component to the HelloStudent component. Two red dashed arrows originate from the `<HelloStudent name = "Thanh"/>` line in the App component. One arrow points to the `{name}` prop in the first HelloStudent function definition, and the other points to the `props.name` prop in the second HelloStudent function definition.

Don't miss

- ◆ The pair of { and } curlies inside of (and) when declaring props

```
1  function Avatar() {
2    return (
3      <img
4        src= "https://i.imgur.com/qj0MgNn.jpeg"
5        alt="Close-up-flowers"
6        width={120}
7        height={200}
8      />
9    );
10 }
11
12 export default function Profile() {
13   return (
14     <Avatar />
15   );
16 }
```

```
function getImageUrl(fileName){
  let imageUrl = 'https://i.imgur.com/' + fileName;
  return imageUrl;
}
```

```
1  function Avatar({ flower, size }) {
2    return (
3      <img
4        src={getImageUrl(flower.fileName)}
5        alt={flower.description}
6        width={size.width}
7        height={size.height}
8        hspace={10} vspace={10}
9      />
10   );
11 }
```

```
31  function Avatar(props) {
32    let flower = props.flower
33    let size = props.size
34    // ...
35 }
```

Profile component



```
return (  
  <>  
    <Avatar flower={flower1} size={{width: 250, height: 300}}/>  
    <Avatar flower={flower2} size={{width: 250, height: 250}}/>  
    <Avatar flower={flower3} size={{width: 250, height: 200}}/>  
  </>  
);
```


Bài tập 01

- ◆ KHÔNG THAY ĐỔI *index.html* và *index.js*
- ◆ Chỉ cập nhật *App.js* và thêm component
- ◆ Tạo file *src/components/Avatar.js* bao gồm các component sau
 - Avatar có tham số (theo gợi ý đề bài)
 - Profile không tham số bao gồm Avatar và các nội dung khác tùy ý
- ◆ Cập nhật *App.js* để
 - Kết xuất ra browser các Avatar khác nhau
 - Kết xuất ra browser các Profile khác nhau

Passing JSX as children

◆ Nest built-in tags

```
<div>
  <img />
</div>
```

◆ Nest components

```
<Card>
  <Avatar />
</Card>
```

```
export default function Profile() {
  return (
    <Card>
      <Avatar
        size={100}
        person={{
          name: 'Katsuko Saruhashi',
          imageId: 'Yfe0qp2'
        }}
      />
    </Card>
  );
}
```

```
function Card({ children }) {
  return (
    <div className="card">
      {children}
    </div>
  );
}
```

```

16 function Toolbar({ onPlayMovie, onUploadImage }) {
17   return (
18     <div>
19       <Button onClick={onPlayMovie} children={'Play Movie'} />
20       <Button onClick={onUploadImage}>Upload Image</Button>
21     </div>
22   );
23 }
24
25 function Button({ onClick, children }) {
26   return (
27     <button onClick={onClick}>
28       {children}
29     </button>
30   );
31 }

```

```

33 function Button1({ onClick, text }) {
34   return (
35     <button onClick={onClick}>
36       {text}
37     </button>
38   );
39 }
40 function Toolbar1({ onPlayMovie, onUploadImage }) {
41   return (
42     <div>
43       <Button1 onClick={onPlayMovie} text={'Play Movie'} />
44       <Button1 onClick={onUploadImage} text={'Upload Image'} />
45     </div>
46   );
47 }

```

Conditional Rendering

Student To-Do List

- Learn React
- Build a Todo App
- Deploy to production

```
1  function Item({ task, isCompleted }) {
2    |   return <li className="item">{task}</li>;
3  }
4
5  export default function ToDoList() {
6    |   return (
7    |     <section>
8    |       <h2 style={{ paddingLeft: 20 }}>Student To-Do List</h2>
9    |       <ul>
10    |         <Item
11    |           isCompleted={true}
12    |           task="Learn React"
13    |         />
14    |         <Item
15    |           isCompleted={true}
16    |           task="Build a Todo App"
17    |         />
18    |         <Item
19    |           isCompleted={false}
20    |           task="Deploy to production"
21    |         />
22    |       </ul>
23    |     </section>
24    |   );
25 }
```

Student To-Do List

- Learn React ✓
- Build a Todo App ✓
- Deploy to production

```
function Item({ task, isCompleted }) {
  if (isCompleted){
    return <li className="item">{task} ✓ </li>;
  }
  return <li className="item">{task}</li>;
}
```

Return null

Student To-Do List

- Learn React ✓
- Build a Todo App ✓
- Deploy to production

Student To-Do List

- Deploy to production

```
1 function Item({ task, isCompleted }) {  
2   if (isCompleted){  
3     // return <li className="item">{task} ✓ </li>;  
4     return null;  
5   }  
6   return <li className="item">{task}</li>;  
7 }
```


Shortcuts in JSX

- ◆ `{cond ? <A /> : <B />}` means “if *cond*, render `<A />`, otherwise `<B />`”.
- ◆ `{cond && <A />}` means “if *cond*, render `<A />`, otherwise nothing”.

```
function Item({ task, isCompleted }) {  
  return <li className="item">{task} {isCompleted && '✅'}</li>;  
}
```

Student To-Do List

- ~~Learn React~~ '✅'
- ~~Build a Todo App~~ '✅'
- Deploy to production

```
function Item({ task, isCompleted }) {  
  return (  
    <li className="item">  
      {isCompleted ? <del>{task} '✅'</del> : task}  
    </li>  
  )  
}
```

Rendering Lists

- ◆ Render components from an array using JavaScript's `map()`
- ◆ Render only specific components using JavaScript's `filter()`

Render a list from ARRAYS

1. Define an array
2. Map the array members into a new array of JSX nodes
3. Return list from component

```
1  const students = [  
2      'Phan Lê Tuấn 207PM27538',  
3      'Nguyễn Xuân Anh 207PM47612',  
4      'Nguyễn Khôi Nguyên 207PM40062',  
5      'Nguyễn Ngọc Tiến 207PM27523',  
6      'Hoàng Văn Thắng 207PM27513'  
7  ];  
8  export default function List() {  
9      const studentList = students.map(student =>  
10         <li>{student}</li>  
11     );  
12     return <ul>{studentList}</ul>;  
13 }
```

- Phan Lê Tuấn 207PM27538
- Nguyễn Xuân Anh 207PM47612
- Nguyễn Khôi Nguyên 207PM40062
- Nguyễn Ngọc Tiến 207PM27523
- Hoàng Văn Thắng 207PM27513

Filtering arrays

1. Create a new array by calling `filter()`
2. Map over filtering array `map()`
3. Return list from component

Examples use *filter()*

- ◆ https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array/filter

```
const words = ['spray', 'elite', 'exuberant', 'destruction', 'present'];  
const result = words.filter((word) => word.length > 6);
```

```
const filteredStudents1 = students.filter(student => student.name === 'Nguyễn Xuân Anh');  
const filteredStudents2 = students.filter(student => student.name.includes('Nguyễn'));
```

```
function filterItems(arr, query) {  
  return arr.filter((el) => el.toLowerCase().includes(query.toLowerCase()));  
}  
const fruits = ["apple", "banana", "grapes", "mango", "orange"];  
console.log(filterItems(fruits, "ap")); // ['apple', 'grapes']  
console.log(filterItems(fruits, "an")); // ['banana', 'mango', 'orange']
```

Bài tập 02

- ◆ Tạo component Profile bao gồm
 - Heading / Title
 - Avatar
 - List không tham số
 - List có tham số
- ◆ Kết xuất ra Browser nhiều Profile

```
theme: {  
  backgroundColor: 'black',  
  color: 'pink'  
}
```

Gregorio Y. Zara's Todos



- Improve the videophone
- Prepare aeronautics lectures
- Work on the alcohol-fuelled engine

Hedy Lamarr's Todos



- Invent new traffic lights
- Rehearse a movie scene
- Improve spectrum technology

Bài tập 03

- ◆ src/assets/data/students.js
- ◆ src/styles/Container.css
- ◆ src/components/Lists.js
- ◆ src/App.js

```
src > assets > data > JS students.js > ...  
1   export const students = [{  
2       |       id: '207PM27538',  
3       |       name: 'Phan Lê Tuấn',  
4       |       image: 'image1.jpg'  
5       |   }, {  
6       |       id: '207PM47612',  
7       |       name: 'Nguyễn Xuân Anh',  
8       |       image: 'image2.jpg'  
9       |   }, {  
10      |       id: '207PM27523',  
11      |       name: 'Nguyễn Ngọc Tiến',  
12      |       image: 'image3.jpg'  
13      |   }  
14  ]
```

Sinh viên tiêu biểu



207PM27538 Phan Lê Tuấn



207PM47612 Nguyễn Xuân Anh



207PM27523 Nguyễn Ngọc Tiến

Bài tập 04

◆ Product

- id: number
- name: string
- price: number
- category: string

1. Hiển thị các sản phẩm có giá (price) trên 60 và thuộc danh mục (category) 'electronics'
2. Hiển thị các sản phẩm thuộc danh mục 'electronics' và 'food' và 'clothing'

Bài tập 05

1. Splitting a list in two
2. Nested lists in one component
3. Extracting a list item component
4. List with a separator

<https://react.dev/learn/rendering-lists>

Display Images

```
1  import React from 'react';
2  import spaghettiImage from '../assets/images/spaghetti.jpg';
3
4  export function DisplayImage() {
5    return <img src={spaghettiImage} alt="Spaghetti" />;
6  }
7
8  export function DisplayImageNoImport() {
9    return <img src={require('../assets/images/spaghetti.jpg')} alt="Spaghetti" />;
10 }
11
12 export function DisplayImageFromPublic() {
13   return (
14     <div>
15       <h1>Truy cập tài nguyên trong thư mục public</h1>
16       <img src={process.env.PUBLIC_URL + 'assets/images/image1.jpg'} alt="Logo"
17         style={{ maxWidth: '10%', padding: '10px'}} />
18     </div>
19   );
20 };
```

- my-app
 - node_modules
 - public
 - assets\images
 - image1.jpg
 - image2.jpg
 - image3.jpg
 - favicon.ico
 - index.html
 - logo192.png
 - logo512.png
 - manifest.json
 - robots.txt
 - src
 - assets\images
 - chicken.jpg
 - spaghetti.jpg
 - components
 - App.css
 - App.js